

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks: 20

Annexure A

Descriptive Written Test

Code of Conduct:

I S. Pooja (192524239) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

S. Pooja
Signature of the student:

level - 4

PS

CSA1714 → Artificial Intelligence 192524237

SLOTD Descriptive Test

1) Smart Home Automation System:

A smart home automation system manages lighting, temperature, and security using real-time inputs such as motion sensors, temperature sensors, and time of day. It also learns user preferences to improve comfort, energy efficiency, and safety. Different types of intelligent agents can be used in this system.

2) Simple Reflex Agent:

★ Works only on the current input using predicted rules.

★ It does not remember previous events.

Example:

★ If motion is detected → Turn on the lights.

★ If no motion for 10 mins → Turn off the lights.

Advantages:

★ Fast Response, easy to implement.

Disadvantages:

★ Cannot learn user habits

★ Cannot make decisions based on past events

3) Model-Based Agents:

★ Maintains an internal model (memory) of the environment.

★ Uses current and previous information to make decisions.

(2)

Example:

Remembers that a room is occupied even if motion temporarily stops.

Advantages:

- ★ Handles partially observable environments.
- ★ More reliable than simple reflex agents.

Disadvantages:

- ★ More complex
- ★ Requires additional memory.

➤ Goal Based Agent:

Makes decision based on achieving specific goals.

Example:

- ★ Automatically closes windows and turns on the AC to maintain the desired temperature.
- ★ Locks all doors when everyone leaves the house.

Advantages:

- ★ Plans actions intelligently.
- ★ Flexible for different situations.

Disadvantages:

Takes more computation time.

Conclusion:

A Model based utility Based Hybrid Agent provides the highest level of comfort, energy efficiency, security & decision making.

(3)

Robot in an Unknown Maze:

A robot is placed in an unknown maze and just find the exit without knowing the layout. Since it has no prior language, uninformed search algorithms such as uniform cost search (UCS) & Iterative Deepening search (IDS) are suitable.

- a) Uniform cost search (UCS): first
- ★ Expands the node with the lowest path cost.
 - ★ Guarantees the least cost path.

Example:

If different paths have different movement cost, UCS chooses the cheapest route to exit.

Advantages:

- ★ Complete (finds a solution if one exists)
- ★ Optimal (finds the lowest cost path)

Disadvantages:

- ★ Requires large memory.
- ★ Can be slow for large search spaces.

Complexity:

$$O(b^l + [C^*/\epsilon])$$

Space Complexity: same as time (high)

b) Iterative Deepening Search:

- ★ Repeatedly performs depth limited search with increasing depth limits until the goal

(4)

is found.

Advantages:

- ★ Uses very little memory.
- ★ Complete for finite search spaces.
- ★ Suitable when solution depth is unknown.

Disadvantages:

- ★ Repeats search from the root.
- ★ Slow due to repeated exploration.

Complexity:

- ★ Time Complexity : $O(b^d)$
- ★ Space Complexity : $O(bd)$

↳ branching factor, $d \rightarrow$ depth of solution

Best:

Effective one is Goal based with uniform cost search (UCS).

Reason:

- ★ The goal based agent focuses on reaching the goal.
- ★ UCS guarantees lowest cost path.

Conclusion:

- ★ Best: Goal Based Agent + Uniform cost Search.
- ★ Alternative: Model Based Agent + Iterative Deepening Search.